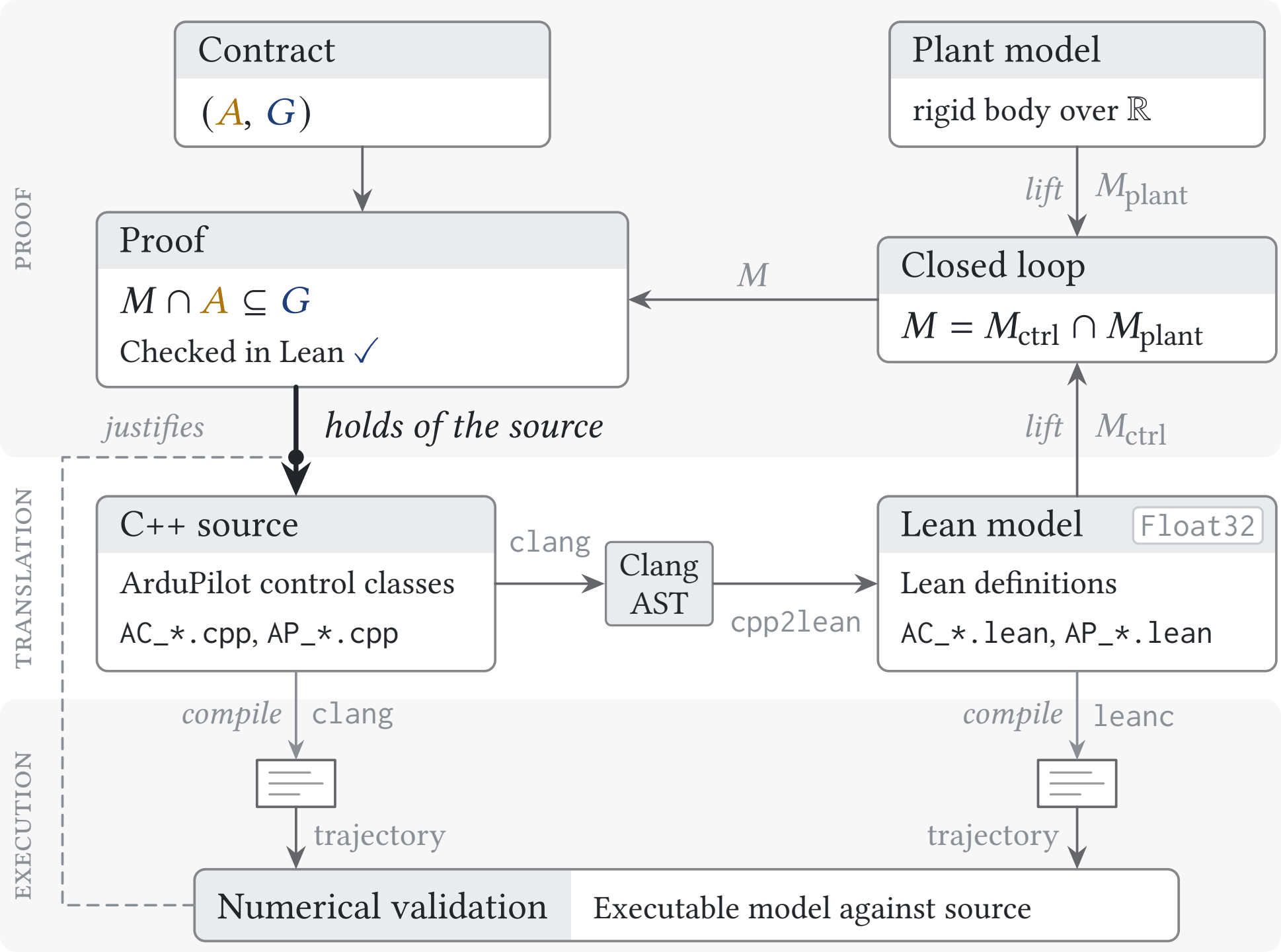

PROOF
TRANSLATION
EXECUTION
Contract
(A, G)
Plant model
rigid body over ℝ
lift
M_plant
Proof
M ∩ A ⊆ G
Checked in Lean ✓
M
Closed loop
M = M_ctrl ∩ M_plant
justifies
holds of the source
lift
M_ctrl
C++ source
ArduPilot control classes
AC_*.cpp, AP_*.cpp
clang
Clang AST
cpp2lean
Lean model
Float32
Lean definitions
AC_*.lean, AP_*.lean
compile
clang
compile
leanc
trajectory
trajectory
Numerical validation
Executable model against source